

Thomas Jiralerspong

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Mila
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Education

Université de Montréal

PhD - Computer Science

In progress

Supervisors: [Yoshua Bengio](#) & [Guillaume Lajoie](#)

Vanier Canada Graduate Scholarship Scholarship (150 000\$)

FRQNT Scholarship (40 000\$) (Rank #1 among all applicants in category)

NSERC Canada Graduate Scholarship (17 500\$)

Hydro-Québec Excellence Scholarship (10 000\$)

Arbour Scholarship (7 500\$)

Massachusetts Institute of Technology

Brains, Minds, and Machines Summer Course

2024

McGill University

B.Sc., Honours Computer Science

2023

Supervisors: [Blake Richards](#) & [Doina Precup](#)

GPA: 4.00/4.00

Exchange semester at the **National University of Singapore**

J.W. McConnell Major Entrance Scholarship (9 000\$)

Research Grants

[Compute Funding for Five Mentored AI Safety Research Projects](#) *2026*

Recipient, Manifund (Manifold for Charity)

US\$390,000

[Exploring the Benefits of Alignment Pretraining](#)

2026

Recipient, Manifund (Manifold for Charity)

US\$83,332

Preprints

Thomas Jiralerspong, Mukesh Ramanathan, Christopher Ackerman, Christina Lu, Guillaume Lajoie, Dan Mossing. "[Affine Anticipation: LLM Answer Representations Are Largely Linearly Predictable from Context.](#)" *Preprint*. 2026.

Refereed Conferences

Thomas Jiralerspong, Trenton Bricken. "Cross-Architecture Model Diffing With Cross-coders." Official Anthropic Blogpost.

Luca Scimeca*, **Thomas Jiralerspong***, Berton Earnshaw, Jason Hartford, Yoshua Bengio. "Learning What Matters: Steering Diffusion via Spectrally Anisotropic Forward Noise." Under review. 2026.

Eric Elmoznino*, **Thomas Jiralerspong***, Yoshua Bengio, Guillaume Lajoie. "A Complexity-Based Theory of Compositionality." In *Forty-Second International Conference on Machine Learning (ICML)*. 2025.

Jin Hwa Lee*, **Thomas Jiralerspong***, Lei Yu, Emily Cheng. “Geometric Signatures of Compositionality Across a Language Model’s Lifetime.” In *The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*. 2025.

Ezekiel Williams, Avery Ryoo*, **Thomas Jiralerspong***, Matt Perich, Guillaume Lajoie. “Expressivity of neural networks with random weights and learned biases.” In *The 13th International Conference on Learned Representations (ICLR)*. 2025.

Jean-Pierre Falet, Hae Beom Lee, Nikolay Malkin, Chen Sun, Dragos Secrieru, **Thomas Jiralerspong**, Dinghuai Zhang, Guillaume Lajoie, Yoshua Bengio. “Delta-AI: Local Objectives for Amortized Inference in Sparse Graphical Models” In *Twelfth International Conference on Learning Representations (ICLR)*. 2024.

Chen Sun, Wannan Yang, **Thomas Jiralerspong**, Dane Malenfant, Benjamin Alsbury-Nealy, Yoshua Bengio, Blake Richards. “Contrastive Retrospection: honing in on critical steps for rapid learning and generalization in RL.” In *Thirty-seventh Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2023.

Flemming Kondrup*, **Thomas Jiralerspong***, Elaine Lau, Nathan de Lara, Jacob Shkrob, My Duc Tran, Doina Precup, Sumana Basu. “Towards Safe Mechanical Ventilation Treatment Using Deep Offline Reinforcement Learning.” In *Thirty-seventh AAAI Conference on Artificial Intelligence (AAAI)*. 2023.

Marshall Wang, John Willes, **Thomas Jiralerspong**, Matin Moezzi. “A Comparison of Classical and Deep Reinforcement Learning Methods for HVAC Control.” In *20th IEEE International Conference on Ubiquitous Intelligence and Computing (UIC)*. 2023.

Refereed Workshops

Samy Mammeri, Rose Nayoung Kwon, Chen Sun, Sabyasachi Sahoo, Christian Gagné, **Thomas Jiralerspong**. “When Does Interleaving Prevent Emergent Misalignment?” In *Mechanistic Interpretability Workshop at ICML*. 2026.

Rose Nayoung Kwon, Samy Mammeri, Sabyasachi Sahoo, Christian Gagné, **Thomas Jiralerspong**. “Hidden-State Similarity Predicts Re-Elicitation After Inoculation Prompting.” In *Mechanistic Interpretability Workshop at ICML*. 2026.

Joachim Schaeffer*, **Thomas Jiralerspong***, Alexander Panfilov*, Roland Zimmerman. “Measuring Control Intervention Awareness Across Frontier LLMs” In *The CAO Workshop at ICLR. Best Paper Award*, Oral Presentation. 2026.

Thomas Jiralerspong*, Flemming Kondrup*, Yoshua Bengio. “Noticing the Watcher: LLM Agents Can Infer CoT Monitoring from Blocking Feedback” In *The ICLR Workshop on Agents in the Wild: Safety, Security, and Beyond*. 2026.

Thomas Jiralerspong, Berton Earnshaw, Jason Hartford, Yoshua Bengio, Luca Scimeca. “Shaping Inductive Bias in Diffusion Models through Frequency-Based Noise Control” In *The ICLR Workshop on Deep Generative Models in Machine Learning: Theory, Principle and Efficacy (DeLTa)*. 2025.

** Equal Contribution

Marco Jiralerspong, **Thomas Jiralerspong**, Vedant Shah, Dhanya Sridhar, Gauthier Gidel. “General Causal Imputation via Synthetic Interventions.” In *The Causal Representation Learning Workshop at NeurIPS*. 2024.

Thomas Jiralerspong*, Xiaoyin Chen*, Yash More, Vedant Shah, Yoshua Bengio. “Efficient Causal Graph Discovery Using Large Language Models.” In *How Far Are We From AGI? Workshop at ICLR*. 2024.

Thomas Jiralerspong*, Flemming Kondrup*, Doina Precup, Khimya Khetarpal. “Forecaster: Towards Temporally Abstract Tree-Search Planning from Pixels.” In *Seventh Workshop on Generalization in Planning at NeurIPS*. 2023.

Flemming Kondrup*, **Thomas Jiralerspong***, Elaine Lau, Nathan de Lara, Jacob Shkrob, My Duc Tran, Doina Precup, Sumana Basu. “Deep Conservative Reinforcement Learning for Personalization of Mechanical Ventilation Treatment.” In *The Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*. 2022.

Research Experience

Astra Fellowship - Mentored by Dan Mossing (Anthropic)

Researcher

Apr 2026 - Present

Project: Characterizing persona space and persona controllability in large language models

Astra Fellowship - Google DeepMind Stream

Researcher

Jan 2026 - Mar 2026

Project: Study of control intervention awareness in large language models

LawZero

Researcher

Jan 2026 - Mar 2026

Project: Studied LLM truth recovery in multi-agent environments with strategically deceptive actors

Anthropic - Mentored by Trenton Bricken

Research Fellow

May 2025 - Oct 2025

Project: Mechanistic interpretability to discover behavioral differences between models

Occam AI

Research Scientist

Jun 2024 - Apr 2025

Projects: Optimization of interactions between network of LLM agents, automated SQL query generation using LLMs

Waabi - Mentored by Kelvin Wong and Chris Zhang

Deep Learning Research Intern

Jun 2023 – Aug 2023

Project: Realistic and controllable traffic simulation using a transformer based variational autoencoder

Reasoning and Learning Lab, Mila/McGill University - Supervised by Prof. Doina Precup

Research Intern

Jan 2022 – Aug 2023

Project: Model-based reinforcement learning with affordance aware tree-search planning directly from pixels

Learning in Neural Circuits Lab, Mila/McGill University - Supervised by Prof. Blake Richards

Research Intern

Sep 2022 – Aug 2023

Project: Contrastive learning to discover critical states for reinforcement learning in sparse reward environments

Vector Institute for A.I. - Mentored by John Willes and Marshall Wang

Machine Learning Research Intern

Sep 2022 – Dec 2022

Project: Model-based reinforcement learning for HVAC control

Project X, Machine Learning Research Competition

Co-leader of McGill's Team

Jun 2021 – Feb 2022

Received the highest score out of 25 submitted papers

Project: Deep offline conservative reinforcement learning for mechanical ventilation treatment

Mentorship

Algoverse AI Research

Principal Investigator

Jun 2026 - Present

Scientific oversight and research direction across 15 AI safety research projects

MARS V (Mentorship for Alignment Research Students) - Meridian Cambridge

Research Mentor

Jun 2026 - Present

Mentorship across 3 AI safety research projects, including work with PhD students (Imperial College London, Polish Academy of Sciences), an ML engineer, and the founder of the Hanoi AI Safety Network

Mila / McGill University

PhD student: Rose Nayoung Kwon (Université Laval)

2026 - Present

PhD student: Aly Kassem (University of Windsor)

2026 - Present

Research intern: Samy Mammeri

2026 - Present

Master's student: Élodie Côté-Gauthier

2023 - 2025

Independent

Undergraduate student: Sohan Venkatesh (Manipal Institute of Technology) *2026 - Present*

Industry Experience

Amazon Web Services (AWS) – S3 Team

Software Development Engineer Intern

May 2022 – Jul 2022

Project: JavaScript/Python tool to automate the Incremental Backup recovery system for AWS S3 (stores ~14 trillion objects)

Square Enix

Software Development Intern

May 2021 – Aug 2021

Project: Localization system to allow a MOBA game to be translated into over 10 languages

Expedia

Software Development Intern

Jun 2019 – Aug 2019

Project: React/TypeScript tool to identify which elements of a webpage are broken and conveniently display them to developers

Teaching

Université de Montréal

Teaching Assistant, Representation Learning 2023

McGill A.I. Society

Organizer/Teaching Assistant, Accelerated Intro to ML 2021 – 2023

McGill University

Teaching Assistant, Software Systems 2021 – 2022

Guest Lecturer, Theory of Machine Learning 2022

Honors

Vanier Canada Graduate Scholarship (150000\$) 2025

FRQNT Master's Scholarship (40000\$) (Rank #1 among all applicants in category) 2024

Arbour Scholarship (7500\$) 2024

Hydro-Québec Excellence Scholarship (10000\$) 2024

Chosen to attend the 10th Heidelberg Laureate Forum 2023

NSERC Canada Graduate Scholarship (17500\$) 2023

University of Montreal Master's Scholarship (5000\$) 2023

McGill Mobility Bursary for Exchanges (6000\$) 2022

Winner of UofT AI's Project X competition (25000\$) 2022

J.W. McConnell Major Entrance Scholarship (9000\$) 2020 – 2022

CIBPA Foundation Bursary (1000\$, 2500\$, 1000\$) 2021, 2022, 2023

Marianopolis College Valedictorian 2020

Governor General of Canada's Academic Medal 2020

Invited
Talks

Canadian Undergraduate Conference on AI (CUCAI) 2022

University of Toronto AI Conference 2022

McGill AI Society Learnathon 2022

Professional
Activities

Mila

Chairman of Lab Representatives 2023 – Present

Chairman of Social Committee 2023 – Present

Executive Member of Recruitment Committee 2023 – Present

McGill AI Society

Senior Advisor 2023 – Present

Technical Project Manager 2021 – 2023

Montreal AI & Neuroscience Conference

Organizer – Introduction to deep learning with PyTorch workshop 2022

	McGill NeuroTech Machine Learning Developer 2021 – 2022
	McGill Robotics Software Developer 2020 – 2021
Languages	Native: English, French Advanced: Italian, Spanish Beginner: Mandarin, Japanese
Skills	Programming Languages: Python, Java, JavaScript, R, C, C++, C#, OCaml, SQL, HTML, CSS Machine Learning Libraries: PyTorch, TensorFlow, Keras, Pandas, NumPy, Matplotlib Other: L ^A T _E X, Slurm, Jupyter Notebooks, Perforce, GitHub, Jira, Unity
Press	Anthropic. Mar 13, 2026. A “Diff” Tool for AI: Finding Behavioral Differences in New Models. Official Anthropic blog post. SciLogs. Nina Beier. Jan 24, 2024. What Do Food and Research Have in Common? More Than You Might Think. The McGill Tribune. Mikaela Shadick. March 15, 2022. Six McGill undergrads win UofT international artificial intelligence competition. McGill Reporter. Richard Deschamps. March 1, 2022. Undergrad team uses machine learning to create a better hospital ventilator.
Advanced Coursework	Université de Montréal Representation Learning Reinforcement Learning & Optimal Control Scaling Laws Causal Inference & Machine Learning Probabilistic Graphical Models McGill University Reinforcement Learning Brain Inspired Artificial Intelligence Honours Math for Machine Learning Probabilistic Programming Network Science National University of Singapore Quantum Computing Information Theory